Exam 2 Prep

Paired Data

Data Loading

```
cholesterol <- read.csv(file = "./data/cholesterol.csv")
colnames(cholesterol)
```

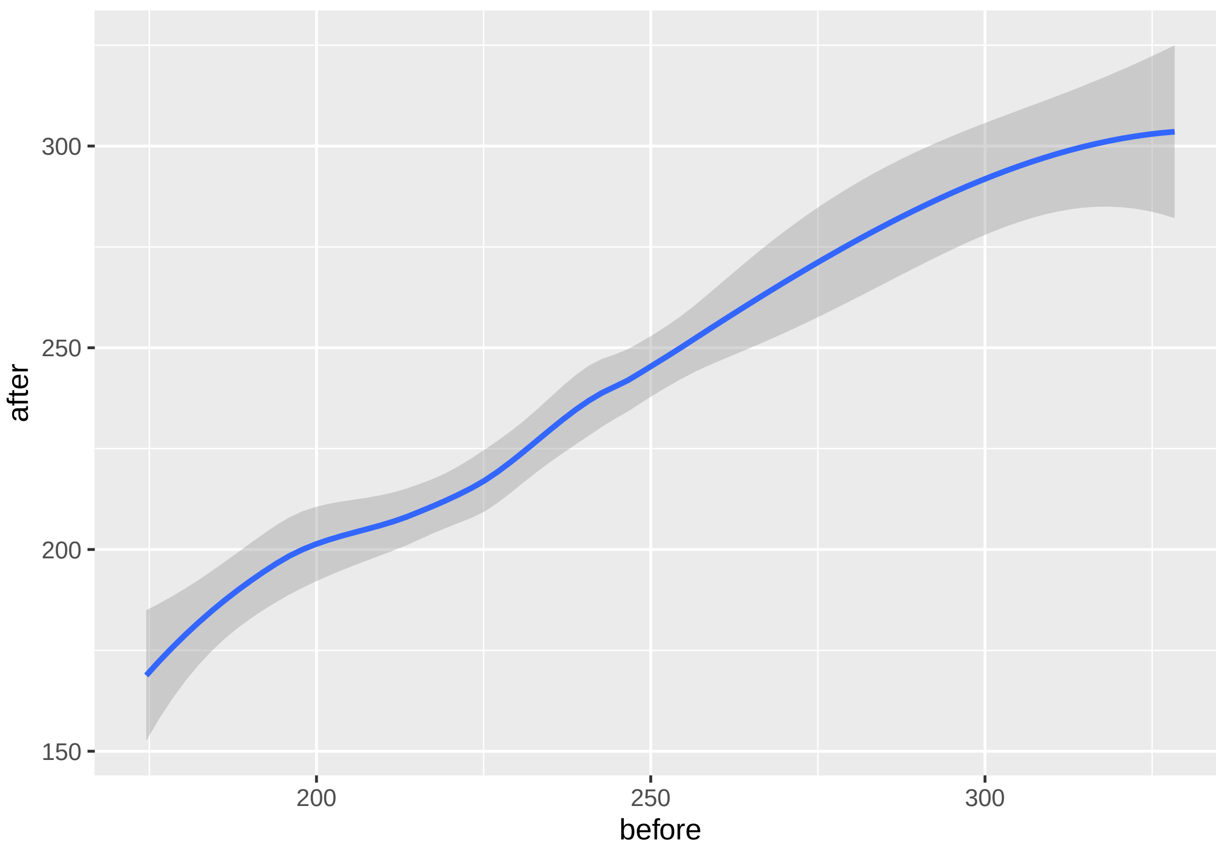
```
## [1] "X"      "before" "after"
```

```
library(ggplot2)
```

Scatterplot

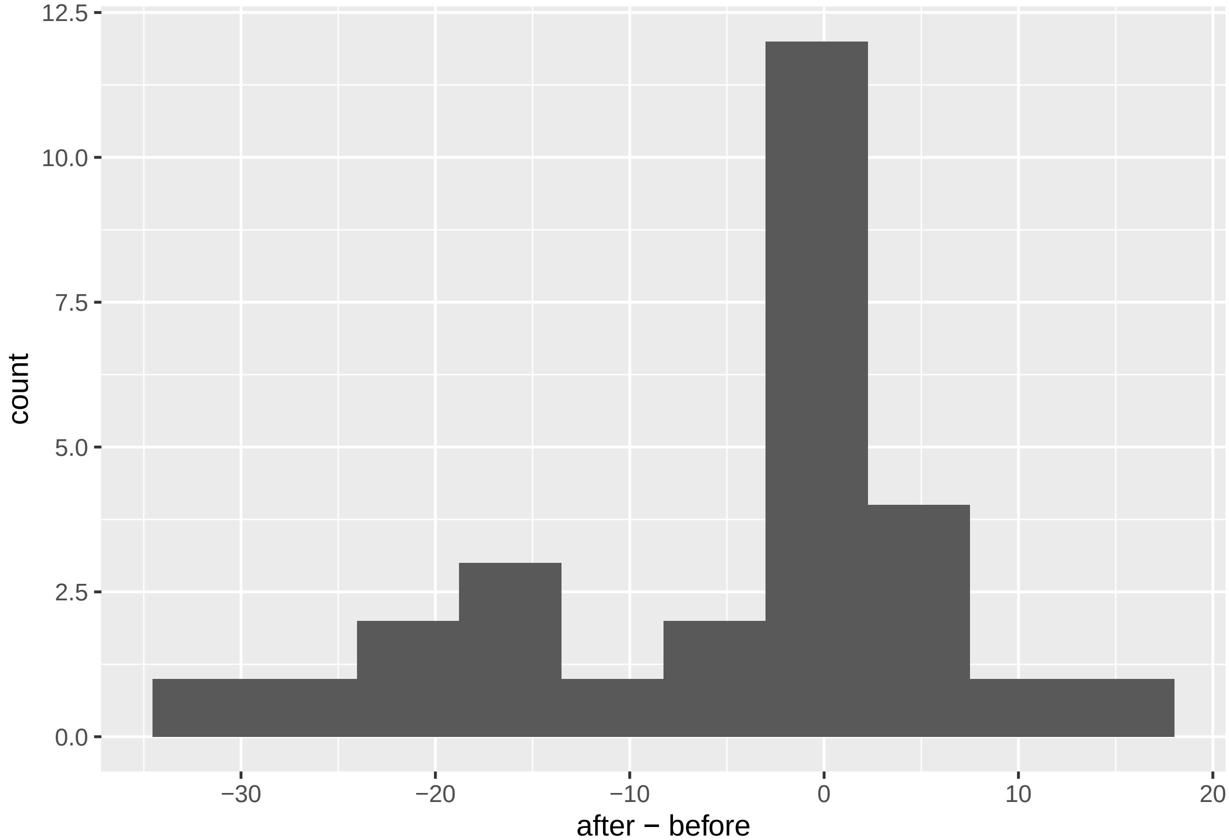
```
ggplot(cholesterol, aes(x = before, y = after)) +
  geom_smooth()
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



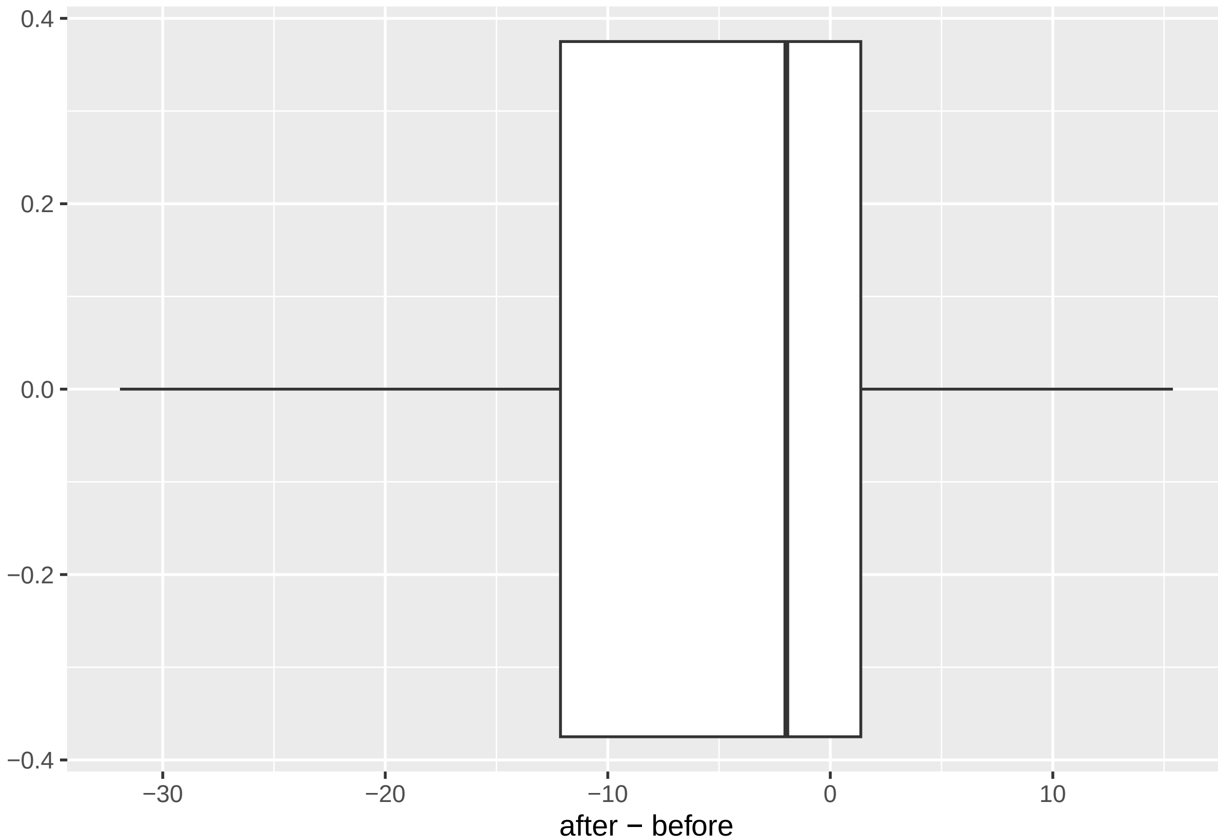
Histogram

```
ggplot(cholesterol, aes(x = after - before)) +
  geom_histogram(bins = 10)
```



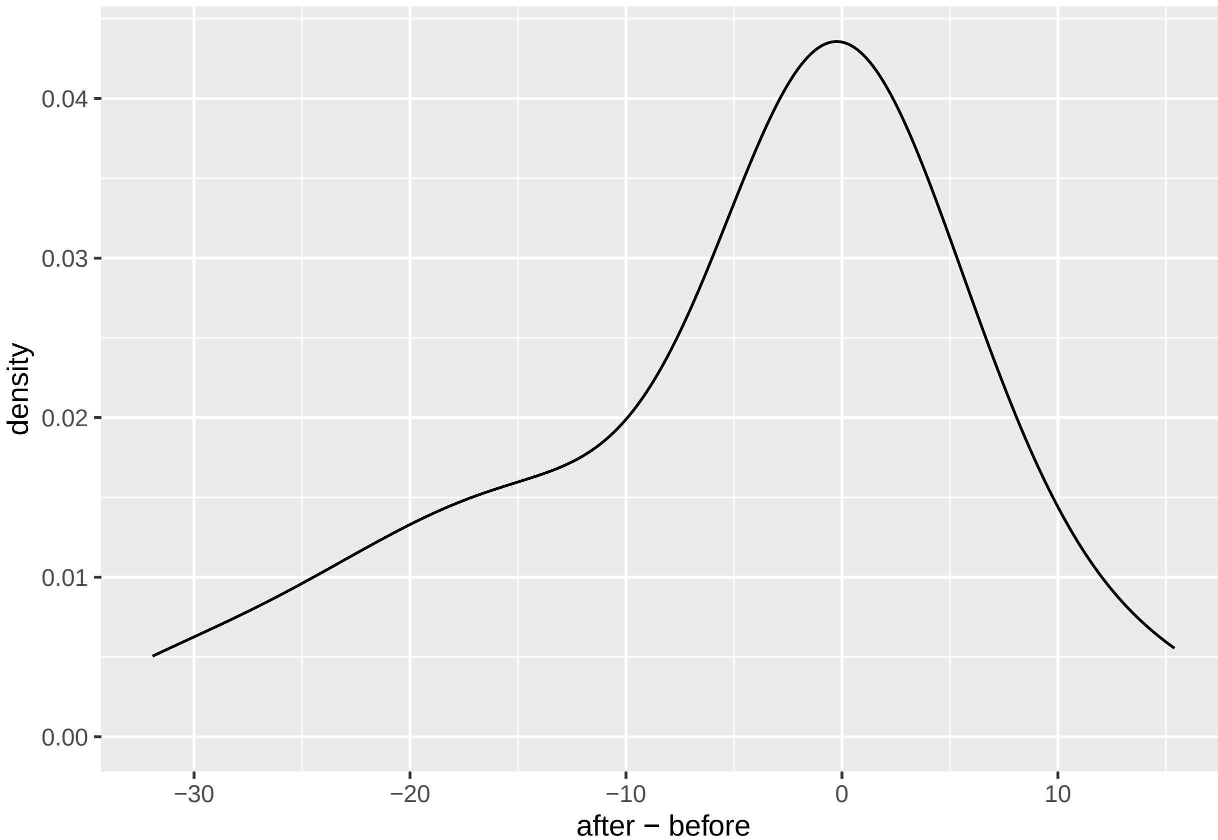
Boxplot

```
ggplot(cholesterol, aes(x = after - before)) +
  geom_boxplot()
```



Density Plot

```
ggplot(cholesterol, aes(x = after - before)) +
  geom_density()
```



Hypothesis Testing

H_0: $\mu_{\text{diff}} = 0$

H_1: $\mu_{\text{diff}} \neq 0$

$t = \frac{\overline{X_{\text{diff}}} - \mu_{\text{diff}}}{s_{\text{diff}}/\sqrt{n}}$

```
xdiff <- mean(cholesterol$after - cholesterol$before)
sdiff <- sd(cholesterol$after - cholesterol$before)
n <- length(cholesterol$after)
mu0 <- 0

alpha <- 0.05

(t <- (xdiff - mu0) / (sdiff / sqrt(n)))
```

```
## [1] -2.373197
```

```
t < alpha
```

```
## [1] TRUE
```